

EE3621: Digital Signal Processing

Lecture 23: Design of FIR filter - windowing method (continued) (III B.Tech EEE)

September 23, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Design** high-rejection FIR filters using Hamming, Blackman, and Kaiser windows.
 2. **Compute** required filter length N using Kaiser empirical design equations.
 3. **Tune** the shape parameter β to trade off transition bandwidth and stopband ripple.
 4. **Compare** performance trade-offs across all standard window families.
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2. MATHEMATICAL FOUNDATIONS

2.1 Standard Window Performance Summary

— Window — Time Formula $w[n]$, $0 \leq n \leq N-1$ — Mainlobe Width $\Delta\omega$ — Peak Sidelobe
— Min Stopband Attenuation A_s — Transition Width Formula — — : — — : — — : — — : —
— : — — : — — — **Rectangular** — 1 — $4\pi/N$ — -13 dB — 21 dB — $\Delta f \approx 0.9/N$ — —
Hann — $0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right)$ — $8\pi/N$ — -31 dB — 44 dB — $\Delta f \approx 3.1/N$ — — **Hamming**
— $0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$ — $8\pi/N$ — -41 dB — 53 dB — $\Delta f \approx 3.3/N$ — — **Blackman** —
 $0.42 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) + 0.08 \cos\left(\frac{4\pi n}{N-1}\right)$ — $12\pi/N$ — -57 dB — 74 dB — $\Delta f \approx 5.5/N$ —

2.2 The Kaiser Window Design Equations

$$w[n] = \frac{I_0\left(\beta \sqrt{1 - \left(\frac{n-\tau}{\tau}\right)^2}\right)}{I_0(\beta)}, \quad \tau = \frac{N-1}{2}$$

Where $I_0(x) = \sum_{k=0}^{\infty} \left[\frac{(x/2)^k}{k!} \right]^2$.

• Shape Parameter β Calculation:

$$\beta = \begin{cases} 0.1102(A_s - 8.7), & \& A_s > 50 \\ 0.5842(A_s - 21)^{0.4} + 0.07886(A_s - 21), & \& 21 \leq A_s \leq 50 \\ 0, & \& A_s < 21 \end{cases}$$

- **Filter Length N Calculation:**

$$N \geq \frac{A_s - 7.95}{14.36 \cdot \Delta f} + 1, \quad \Delta f = \frac{\omega_s - \omega_p}{2\pi}$$

Pictorial Representations

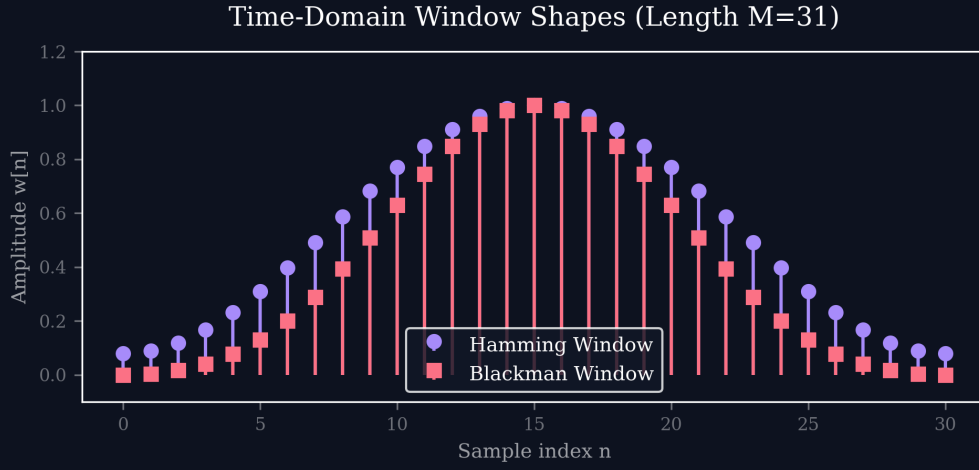


Figure 1: Window Comparison Shapes

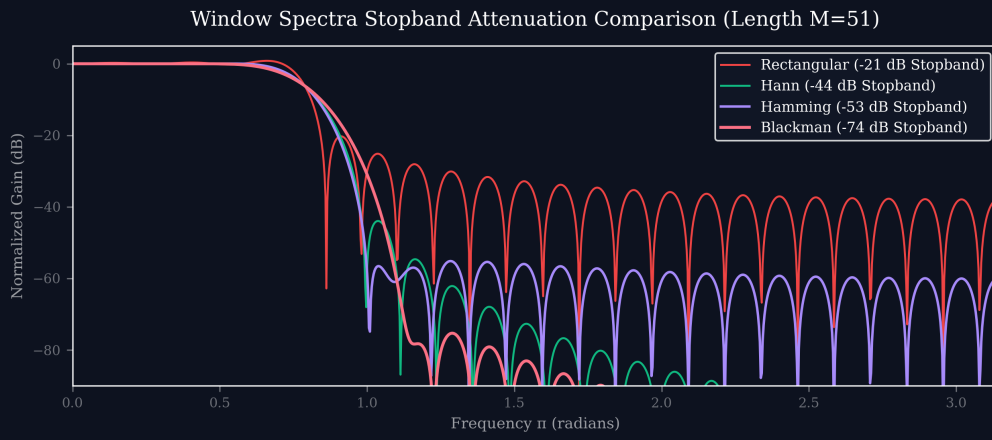


Figure 2: Sidelobe Suppression